GRADE 10 CHEMISTRY

LEARNER'S NOTES

Comprehensive notes covering Inorganic Chemistry and Physical Chemistry, based on the KICD Senior School Curriculum Design

Strands covered:

1. Inorganic Chemistry
2. Physical Chemistry

NOTES

HOW TO USE THESE NOTES

Welcome to your Grade 10 Chemistry Notes! This book has been written specifically for you, a Senior School learner aged between 14 and 16 years, to help you understand and enjoy Chemistry.

These notes are organised into seven topics, following the official curriculum design. Each topic contains the following features to support your learning:

- Learning Outcomes – what you should be able to do by the end of the topic.
- 💡 Idea Box – interesting facts and important points to remember.
- 🔑 Key Terms – definitions of important vocabulary.
- ⚠️ Safety First – important laboratory safety reminders.
- 🤔 Reflect and Discuss – open questions to think about and discuss with your classmates and teacher. There are no single 'correct' answers to these questions — they are meant to help you think more deeply.
- 📖 Case Study – real-life examples showing how Chemistry applies to our world.
- Worksheets – practical and project activities to deepen your understanding.
- Exercises – questions to check your understanding as you finish each section.
- End of Topic Questions – more challenging questions to test what you have learned in the whole topic.

All answers to the Exercises and End of Topic Questions are provided in the Answer Key at the end of this book (Chapter 8). Try to answer every question on your own or with your classmates before checking the answer key — this is how you will learn best!

💡 IDEA BOX: A Note to the Learner

Chemistry is best learned by doing. Take part actively in all laboratory activities, ask questions, and always link what you learn in class to the world around you. Enjoy your journey into the fascinating world of Chemistry!

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TOPIC 1: INTRODUCTION TO CHEMISTRY

By the end of this topic, you should be able to: explain the meaning of Chemistry as a field of science; explore the role of Chemistry in day to day life; examine the effects of drug and substance use; and promote the rights and responsibilities to a safe and healthy learning environment.

1.1 What is Chemistry?

Chemistry is the branch of science that studies the composition, structure, properties and changes of matter. Matter is anything that has mass and occupies space. Every object around you — the air you breathe, the food you eat, your exercise book, your own body — is made of matter, and Chemistry helps us understand what these things are made of and how they behave and change.

Chemistry is often called the 'central science' because it links Physics, Biology, Geology, Agriculture and many other fields together.

🔑 **KEY TERMS**

Chemistry: the scientific study of matter, its properties, and the changes it undergoes.

Matter: anything that has mass and occupies space.

Element: a pure substance that cannot be broken down into simpler substances by chemical means.

Compound: a substance formed when two or more elements chemically combine in fixed proportions.

1.2 Branches of Chemistry

Chemistry is a broad subject that is divided into several branches, each focusing on a different aspect of matter:

- Organic Chemistry – the study of carbon-containing compounds (e.g. fuels, plastics, medicines).
- Inorganic Chemistry – the study of non-carbon compounds, elements, minerals and metals.
- Physical Chemistry – the study of the physical properties of matter and the energy changes that accompany reactions.
- Analytical Chemistry – concerned with identifying and measuring the composition of substances.
- Biochemistry – the chemistry of living organisms.
- Nuclear Chemistry – the study of reactions taking place in the nucleus of an atom.

1.3 Careers Related to Chemistry

A good foundation in Chemistry opens doors to many careers, including:

- Medicine, Pharmacy and Nursing
- Agricultural science and food technology
- Petroleum and mining engineering
- Forensic science
- Environmental science
- Manufacturing and quality control

- Teaching and research

Note: Careers in Chemistry are open to everyone regardless of gender. Historically, some careers in science have been wrongly stereotyped as being 'for boys'. Girls and boys should be equally encouraged to pursue any career path they are interested in.

🤔 REFLECT AND DISCUSS

1. Name three things you used today that are products of the chemical industry.
2. Why do you think some people believe that science careers are 'for boys only'? Do you agree with this view? Explain.
3. Which branch of Chemistry interests you the most, and why?

1.4 Chemistry in Our Daily Lives

Chemistry touches almost every activity of our daily lives:

- Agriculture – manufacture of fertilisers and pesticides to improve crop yield.
- Pharmaceutical industry – development of drugs and vaccines.
- Medicine – diagnosis and treatment of diseases.
- Manufacturing industry – production of plastics, textiles, soap and detergents.
- Food industry – food preservation, flavouring and packaging.
- Entertainment industry – manufacture of dyes, cosmetics and fireworks.
- Energy sector – production and refining of fuels, and development of batteries and solar cells.

💡 IDEA BOX: Chemistry Around You

The toothpaste you use, the soap you bathe with, the charcoal you cook with, and even the paper in this book are all products of chemical processes!

1.5 Effects of Drug and Substance Use

A drug is any substance that, when taken into the body, alters one or more of its functions. Some drugs are prescribed by qualified health workers to treat illness (e.g. antibiotics, painkillers), while others are abused for non-medical reasons (e.g. alcohol, tobacco, bhang, miraa).

Prescription and dosage: A prescription is an instruction given by a qualified health worker on the correct drug and dosage (amount and frequency) to be taken. Taking drugs without prescription, or in the wrong dosage, is dangerous and is called drug abuse.

Effects of drug and substance abuse include:

- Poor academic performance and loss of concentration
- Damage to body organs such as the liver, kidneys and brain
- Addiction and dependency
- Risky behaviour that can lead to accidents, injury or death
- Breakdown of family and social relationships

⚠️ SAFETY FIRST

Never take any medicine without the guidance of a qualified health worker.

Always follow the dosage and instructions written on the drug label.

Report any case of drug and substance abuse to a trusted teacher, parent or guardian.

1.6 Consumer Rights and a Safe Learning Environment

As consumers, we have the right to accurate information on any product we use. Always check product labels for the following before use:

- Quality marks / certification marks (e.g. the KEBS diamond mark)
- Expiry and manufacture dates
- Nutritional information
- Preservatives used
- Cautionary/warning messages

Every learner also has the right to a safe and healthy learning environment, especially in the laboratory. This includes the responsibility to follow all safety rules, use protective equipment, and report any hazards immediately.

📖 CASE STUDY: Reading Product Labels

Amina wanted to buy juice for her younger sister. At the shop, she found two brands of juice at almost the same price. She checked the labels: one had a clear expiry date, a KEBS quality mark and listed all preservatives used; the other had no expiry date visible and no quality mark. Amina chose the first brand and explained to the shopkeeper why consumer information matters.

This shows why every consumer, however young, should always check labels before purchasing products — it is both a right and a responsibility.

Topic 1 Worksheet: Project Activity

In groups of four to six learners, design a poster to sensitise your school community on the risks of drug and substance abuse. Your poster should include:

1. A catchy title
2. At least three effects of drug and substance abuse
3. An illustration or diagram
4. A message of hope or a helpline/contact for support

Topic 1 Exercise

1. Define the term 'Chemistry'.
2. List any four branches of Chemistry and state what each branch studies.
3. State three careers that require knowledge of Chemistry.

4. Explain the meaning of the term 'drug abuse'.
5. Give three pieces of information you should check on a product label before buying it.

Topic 1: End of Topic Questions

1. Chemistry has been described as the 'central science'. Explain what this statement means.
2. Describe how Chemistry is applied in the agricultural sector.
3. Outline four effects of drug and substance abuse on a learner's life.
4. Explain the meaning of a 'prescription' and why it is important to follow it strictly.
5. Discuss two rights and two responsibilities of a learner in ensuring a safe school laboratory environment.

TOPIC 2: THE ATOM

By the end of this topic, you should be able to: describe the structure of the atom; determine the relative atomic mass of elements; write the electron arrangement of elements using s and p notation; and develop interest in the study of the structure of the atom.

2.1 The Structure of the Atom

An atom is the smallest particle of an element that can take part in a chemical reaction. Atoms are made up of three main sub-atomic particles:

Sub-atomic Particle	Relative Charge	Relative Mass	Location
Proton	+1	1	Nucleus
Neutron	0	1	Nucleus
Electron	-1	1/1840 (negligible)	Orbiting the nucleus (energy levels)

The nucleus is the tiny, dense centre of the atom containing protons and neutrons. Electrons move around the nucleus in regions called energy levels (or shells).

A Brief History of Atomic Models

- John Dalton (1803): proposed that atoms are tiny, indivisible, indestructible solid spheres — the 'billiard ball' model.
- J.J. Thomson (1897): discovered the electron and proposed the 'plum pudding' model — a sphere of positive charge with electrons embedded in it.
- Ernest Rutherford (1911): through his gold foil experiment, discovered that an atom has a small, dense, positively charged nucleus with electrons orbiting around it, and that most of the atom is empty space.
- Niels Bohr (1913): proposed that electrons move in fixed circular energy levels (shells) around the nucleus.

💡 IDEA BOX: The Gold Foil Experiment

Rutherford fired tiny positively-charged particles at a very thin sheet of gold foil. Most particles passed straight through, but a few bounced back. This surprising result showed that atoms are mostly empty space with a small, dense, positively charged nucleus at the centre.

🤔 REFLECT AND DISCUSS

1. Why do you think scientific models of the atom have changed over time rather than being fixed from the start?
2. What does the Rutherford experiment teach us about how scientists develop and test theories?

2.2 Atomic Number, Mass Number and Number of Particles

The atomic number (Z) of an element is the number of protons in the nucleus of its atom. It also equals the number of electrons in a neutral atom.

The mass number (A) is the total number of protons and neutrons in the nucleus (also called nucleons).

Number of neutrons = Mass number (A) − Atomic number (Z)

Example: Sodium has atomic number 11 and mass number 23. Number of protons = 11, number of electrons = 11, number of neutrons = 23 − 11 = 12.

2.3 Isotopes and Relative Atomic Mass

Isotopes are atoms of the same element that have the same atomic number (same number of protons) but different mass numbers (different number of neutrons). Isotopes of an element have identical chemical properties but slightly different physical properties, such as mass.

Example: Chlorine has two common isotopes, chlorine-35 (75% abundance) and chlorine-37 (25% abundance).

The Relative Atomic Mass (R.A.M.) of an element is the weighted average mass of the isotopes of that element compared to 1/12 the mass of a carbon-12 atom. It is calculated using the formula:

R.A.M. = [(mass of isotope 1 × % abundance) + (mass of isotope 2 × % abundance) + ...] ÷ 100

Worked Example: Calculate the R.A.M. of chlorine given that it has 75% chlorine-35 and 25% chlorine-37.

R.A.M. = [(35 × 75) + (37 × 25)] ÷ 100 = [2625 + 925] ÷ 100 = 3550 ÷ 100 = 35.5

📖 CASE STUDY: Isotopes in Medicine and Archaeology

Isotopes are not just theoretical — they are used in real life! Carbon-14, a radioactive isotope of carbon, is used by archaeologists and historians to determine the age of ancient plant and animal remains through a process called radiocarbon dating. In hospitals, radioactive isotopes such as Iodine-131 are used to diagnose and treat thyroid conditions.

2.4 Energy Levels and Orbitals

Electrons in an atom occupy regions of space around the nucleus called energy levels (or shells), numbered 1, 2, 3, 4 starting from the level closest to the nucleus. Each energy level can hold a maximum number of electrons, given by the formula $2n^2$, where n is the energy level number.

Energy Level (n)	Maximum electrons ($2n^2$)
1	2
2	8
3	18 (only 8 used up to Ca in this course)
4	2 (only s-sublevel used up to Ca)

Within each energy level, electrons occupy smaller regions called orbitals, which are grouped into sub-levels labelled s, p, d and f. In this course, we will focus on s and p orbitals only, for the first 20 elements (Hydrogen to Calcium).

- An s orbital can hold a maximum of 2 electrons.
- A p sub-level has three p orbitals and can hold a maximum of 6 electrons in total.
- Electrons fill the lowest energy orbitals first before occupying higher energy ones (the Aufbau principle).

2.5 Electron Arrangement Using s and p Notation (Elements 1–20)

The electron arrangement of an atom shows how its electrons are distributed in the s and p sub-levels. The order of filling is: 1s, 2s, 2p, 3s, 3p, 4s.

Element	Symbol	Atomic Number	Electron Arrangement
Hydrogen	H	1	$1s^1$
Helium	He	2	$1s^2$
Lithium	Li	3	$1s^2\ 2s^1$
Beryllium	Be	4	$1s^2\ 2s^2$
Boron	B	5	$1s^2\ 2s^2\ 2p^1$
Carbon	C	6	$1s^2\ 2s^2\ 2p^2$
Nitrogen	N	7	$1s^2\ 2s^2\ 2p^3$
Oxygen	O	8	$1s^2\ 2s^2\ 2p^4$
Fluorine	F	9	$1s^2\ 2s^2\ 2p^5$
Neon	Ne	10	$1s^2\ 2s^2\ 2p^6$
Sodium	Na	11	$1s^2\ 2s^2\ 2p^6\ 3s^1$
Magnesium	Mg	12	$1s^2\ 2s^2\ 2p^6\ 3s^2$
Aluminium	Al	13	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$
Silicon	Si	14	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$
Phosphorus	P	15	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$
Sulphur	S	16	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$
Chlorine	Cl	17	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$
Argon	Ar	18	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$
Potassium	K	19	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$
Calcium	Ca	20	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$

💡 IDEA BOX: Notice the Pattern

Notice that potassium (K) and calcium (Ca) start filling the 4s sub-level before the 3p sub-level is completely used up in 3d — this is because, at this stage, the 4s sub-level is slightly lower in energy than 3d. You will learn more about the 3d sub-level in later topics.

🤔 REFLECT AND DISCUSS

1. Why do you think electrons fill the lowest available energy levels first rather than jumping to higher ones?
2. Working with a partner, model the electron arrangement of oxygen using locally available materials such as bottle tops or seeds. Explain your model to the class.

Topic 2 Worksheet

Using dot-and-cross or shell diagrams and locally available materials (bottle tops, seeds, stones), model the atomic structure of any three elements from Hydrogen to Calcium. For each model, indicate the number of protons, neutrons and electrons, and the electron arrangement in s and p notation.

Topic 2 Exercise

1. Define the terms: (a) atomic number, (b) mass number, (c) isotope.
2. An atom of magnesium has atomic number 12 and mass number 24. State the number of protons, neutrons and electrons in this atom.
3. Bromine exists as two isotopes: bromine-79 (50.7% abundance) and bromine-81 (49.3% abundance). Calculate the relative atomic mass of bromine to 1 decimal place.
4. Write the electron arrangement of the following elements using s and p notation: (a) Nitrogen (Z=7) (b) Silicon (Z=14) (c) Potassium (Z=19).
5. State the maximum number of electrons that can occupy: (a) an s orbital (b) a p sub-level.

Topic 2: End of Topic Questions

1. Describe, in the correct order, the contributions of Dalton, Thomson, Rutherford and Bohr to the development of the atomic model.
2. Explain why isotopes of the same element have identical chemical properties.
3. Neon has three isotopes: neon-20 (90.5%), neon-21 (0.3%) and neon-22 (9.2%). Calculate its relative atomic mass to one decimal place.
4. Draw and label a simple diagram showing the structure of a lithium atom, including the nucleus and energy levels.
5. Write the full electron arrangement, in s and p notation, of an element with atomic number 16. Name the element.

TOPIC 3: THE PERIODIC TABLE

By the end of this topic, you should be able to: relate the position of an element in the periodic table to its electron arrangement; illustrate ion formation of elements; derive the formulae of compounds; and write balanced equations for chemical reactions.

3.1 Development of the Periodic Table

Early scientists tried to arrange the known elements in a way that showed patterns in their properties. In 1869, the Russian chemist Dmitri Mendeleev arranged the elements known at the time in order of increasing atomic mass, and noticed that similar properties recurred at regular intervals (periodicity). He even left gaps for elements that had not yet been discovered, and successfully predicted their properties.

In 1913, Henry Moseley showed that elements should instead be arranged in order of increasing atomic number, which corrected a few inconsistencies in Mendeleev's table. This is the basis of the Modern Periodic Table used today.

💡 IDEA BOX: Mendeleev's Genius

Mendeleev left gaps in his table for elements he believed existed but had not yet been discovered, such as 'eka-silicon'. Years later, the element germanium was discovered with almost exactly the properties Mendeleev had predicted!

3.2 Groups, Periods and Electron Arrangement

The Modern Periodic Table arranges elements in horizontal rows called periods and vertical columns called groups.

- A period shows elements with the same number of occupied energy levels (shells).
- A group shows elements with the same number of electrons in the outermost (valence) shell, and therefore similar chemical properties.

For example, Sodium (2,8,1) and Potassium (2,8,8,1) are both in Group I because they each have 1 electron in their outermost shell. Sodium is in Period 3 (3 shells) while Potassium is in Period 4 (4 shells).

Group	Common Name	Valence Electrons	Examples
I	Alkali metals	1	Li, Na, K
II	Alkaline earth metals	2	Mg, Ca
VII	Halogens	7	F, Cl, Br
VIII (0)	Noble gases	8 (2 for He)	He, Ne, Ar

Elements found between Group II and Group III in Periods 4–7 are called transition elements. You will learn more about them in later grades.

3.3 Ion Formation

Atoms react in order to attain a stable, full outer-shell electron arrangement, similar to that of the nearest noble gas (the octet rule, or duplet rule for elements near helium). Atoms achieve this stability by losing, gaining or sharing electrons.

- Metal atoms tend to lose electrons from their outer shell to form positively charged ions called cations. Example: $Na \rightarrow Na^{+} + e^{-}$

- Non-metal atoms tend to gain electrons to fill their outer shell, forming negatively charged ions called anions. Example: $Cl + e^- \rightarrow Cl^-$

The charge on an ion equals the number of electrons lost (positive) or gained (negative).

🔑 **KEY TERMS**

Cation: a positively charged ion, formed when an atom loses one or more electrons.

Anion: a negatively charged ion, formed when an atom gains one or more electrons.

Valency: the combining power of an atom, equal to the number of electrons it loses, gains or shares to attain stability.

Oxidation number: a number assigned to an atom or ion that indicates its degree of oxidation and the charge it would have.

3.4 Valency, Oxidation Numbers and Radicals

The valency of a simple ion is usually the same as its charge. For most Group I, II and III metals and Group V, VI, VII non-metals, the valency can be predicted directly from the group number:

Group	I	II	III	V	VI	VII
Valency of ion formed	1	2	3	3	2	1

Some elements, especially transition metals, show variable oxidation numbers. For example, iron can form Fe^{2+} (iron(II)) or Fe^{3+} (iron(III)), and copper can form Cu^{+} (copper(I)) or Cu^{2+} (copper(II)).

A radical (or polyatomic ion) is a group of atoms that are chemically bonded together and carry an overall charge, behaving as a single ion in reactions.

Radical	Formula	Charge
Hydroxide	OH	-1
Nitrate	NO_3	-1
Carbonate	CO_3	-2
Sulphate	SO_4	-2
Ammonium	NH_4	+1

3.5 Writing Formulae of Compounds

To write the formula of a compound, cross over the numerical values of the valencies (or charges) of the combining ions, ignoring their signs, and simplify where possible.

Worked Example 1: Write the formula for magnesium chloride. Magnesium ion is Mg^{2+}, chloride ion is Cl^-. Crossing over the valencies gives $MgCl_2$.

Worked Example 2: Write the formula for aluminium sulphate. Aluminium ion is Al^{3+}, sulphate ion is SO_4^{2-}. Crossing over gives $Al_2(SO_4)_3$.

🤔 **REFLECT AND DISCUSS**

1. Why do you think brackets are used around a radical when more than one of that radical is present in a formula, e.g.

$Al_2(SO_4)_3$?

2. In your own words, explain why atoms 'want' to lose, gain or share electrons.

3.6 Writing and Balancing Chemical Equations

A chemical equation is a shorthand way of representing a chemical reaction using symbols and formulae. A balanced equation has equal numbers of atoms of each element on both the reactant and product sides, in line with the Law of Conservation of Mass.

Steps to balance an equation:

1. Write the word equation first.
2. Write the correct formulae of all reactants and products.
3. Balance the number of atoms of each element by adjusting the numbers (coefficients) in front of formulae — never change a subscript within a formula.
4. Check that both sides have equal numbers of each type of atom.

Example: Balance the equation for the reaction between magnesium and hydrochloric acid.

Unbalanced: $Mg + HCl \rightarrow MgCl_2 + H_2$

Balanced: $Mg + 2HCl \rightarrow MgCl_2 + H_2$

📖 CASE STUDY: Balancing Equations in Industry

Balanced chemical equations are not just an exam requirement — industries rely on them to calculate the exact amounts of raw materials needed for large-scale production. For example, fertiliser companies use balanced equations to calculate how much ammonia and nitric acid are needed to produce a required mass of ammonium nitrate fertiliser, minimising waste and cost.

Topic 3 Worksheet

Using the periodic table provided by your teacher, complete the following table for the elements: Na, Mg, Al, O, F.

Element	Group	Period	Electron Arrangement	Ion Formed	Valency

Topic 3 Exercise

1. Explain the difference between a group and a period in the periodic table.
2. Write the formulae of the following compounds: (a) calcium chloride (b) sodium sulphate (c) ammonium nitrate.

3. Draw dot-and-cross diagrams to show the formation of magnesium oxide from magnesium and oxygen atoms.
4. Balance the following equations: (a) $Na + O_2 \rightarrow Na_2O$ (b) $CaCO_3 \rightarrow CaO + CO_2$ (c) $Fe + Cl_2 \rightarrow FeCl_3$.
5. State the oxidation numbers of iron in $FeCl_2$ and $FeCl_3$.

Topic 3: End of Topic Questions

1. Describe how Mendeleev arranged elements in his periodic table and state one weakness that Moseley later corrected.
2. An element X has the electron arrangement 2,8,2. State (a) its group (b) its period (c) the ion it is likely to form.
3. Write balanced chemical equations for: (a) the reaction between zinc and dilute hydrochloric acid, producing zinc chloride and hydrogen gas. (b) the reaction between sodium hydroxide and sulphuric acid, producing sodium sulphate and water.
4. Explain why elements in the same group of the periodic table have similar chemical properties.
5. Write the formula for iron(III) sulphate, showing your working.

TOPIC 4: CHEMICAL BONDING

By the end of this topic, you should be able to: illustrate bond types in elements, molecules and compounds; investigate the relationship between bond types and physical properties; relate bond types and resultant structures to uses of substances; and appreciate the uses of different substances based on their bonding.

4.1 Why Atoms Bond

Atoms bond with one another in order to attain a stable electron arrangement, usually resembling that of the nearest noble gas (a full outer shell — the octet rule, or 2 electrons for elements near helium, the duplet rule). This can happen by transferring electrons completely (ionic bonding) or by sharing electrons (covalent bonding), or through a 'sea' of shared electrons in metals (metallic bonding).

4.2 Ionic (Electrovalent) Bonding

An ionic bond is the strong electrostatic force of attraction between oppositely charged ions, formed when electrons are completely transferred from a metal atom to a non-metal atom.

Example: Formation of sodium chloride (NaCl). Sodium (2,8,1) loses one electron to chlorine (2,8,7), forming Na^+ (2,8) and Cl^- (2,8,8). The oppositely charged ions attract each other strongly, forming a giant ionic lattice.

💡 IDEA BOX: Giant Ionic Lattices

Ionic compounds do not exist as single 'molecules'. Instead, billions of positive and negative ions pack together in a repeating three-dimensional pattern called a giant ionic lattice, held together by strong electrostatic forces in all directions.

4.3 Covalent Bonding

A covalent bond is formed when two non-metal atoms share one or more pairs of electrons so that both atoms attain a stable electron arrangement.

- A single covalent bond involves sharing one pair of electrons, e.g. in H_2 or HCl.
- A double covalent bond involves sharing two pairs of electrons, e.g. in O_2 or CO_2.
- A triple covalent bond involves sharing three pairs of electrons, e.g. in N_2.

Dative (Coordinate) Covalent Bonding: this is a special type of covalent bond in which both shared electrons come from the same atom (the donor), while the other atom (the acceptor) contributes none. Example: the ammonium ion (NH_4^+) is formed when the lone pair of electrons on the nitrogen atom in ammonia (NH_3) is donated to a hydrogen ion (H^+).

4.4 Metallic Bonding

A metallic bond is the strong force of attraction between positively charged metal ions and a 'sea' of delocalised (free-moving) electrons that surround them. This explains why metals conduct electricity and heat well, and why they can be hammered into shape (malleable) or drawn into wires (ductile) without breaking, since the layers of ions can slide over one another while remaining held together by the sea of electrons.

4.5 Intermolecular Forces

Besides the strong bonds within molecules (covalent bonds), there are weaker forces of attraction between separate molecules, called intermolecular forces. These determine properties such as melting and boiling points.

- Van der Waals forces: weak, temporary forces of attraction that exist between all molecules, arising from temporary shifts in electron distribution. They are the weakest intermolecular force and increase with molecular size.
- Hydrogen bonding: a stronger intermolecular force that occurs between molecules where hydrogen is directly bonded to a small, highly electronegative atom such as nitrogen, oxygen or fluorine (e.g. in water, H_2O, or ammonia, NH_3). This is why water has an unusually high boiling point for such a small molecule.

🤔 REFLECT AND DISCUSS

1. Why do you think water (H_2O), a very small molecule, has a much higher boiling point than expected compared to other similar-sized molecules?
2. Working in groups, use locally available materials (e.g. beads, sticks, modelling clay) to build a model showing ionic bonding in sodium chloride and covalent bonding in carbon dioxide. Present your models to the class.

4.6 Bond Type, Structure and Physical Properties

The type of bonding and structure of a substance strongly determines its physical properties, such as melting point, boiling point, solubility, and electrical conductivity.

Structure Type	Example	Melting/Boiling Point	Electrical Conductivity	Solubility in Water
Giant ionic	Sodium chloride, NaCl	High	Conducts when molten or in solution (not solid)	Usually soluble
Simple molecular	Iodine, CO_2, water	Low	Does not conduct	Varies; often insoluble unless polar
Giant covalent (atomic)	Diamond, Graphite, SiO_2	Very high	Diamond/SiO_2: does not conduct. Graphite: conducts	Insoluble
Giant metallic	Aluminium, iron, copper	High	Conducts (solid and molten)	Insoluble

📖 CASE STUDY: Diamond and Graphite — Same Element, Different Properties

Diamond and graphite are both made purely of carbon atoms, yet they have very different properties. In diamond, each carbon atom forms four strong covalent bonds to other carbon atoms in a rigid three-dimensional giant structure, making diamond the hardest known natural substance — used in cutting tools and jewellery.

In graphite, each carbon atom forms only three covalent bonds, creating flat layers of hexagonal rings. The layers are held together by weak Van der Waals forces, allowing them to slide over each other easily — this is why graphite is soft and slippery, and is used as a lubricant and in pencils. The free (delocalised) electron from each carbon atom also allows graphite to conduct electricity, unlike diamond.

4.7 Uses of Substances Based on Bond Type and Structure

Substance	Bond/Structure Type	Use	Reason
Diamond	Giant covalent	Cutting tools, drill tips	Extremely hard
Graphite	Giant covalent (layered)	Pencils, lubricants, electrodes	Soft, slippery, conducts electricity
Sodium chloride	Giant ionic	Food seasoning, industrial raw material	Soluble, forms ions in solution
Copper	Giant metallic	Electrical wiring	Good conductor, ductile
Aluminium	Giant metallic	Aircraft bodies, cooking utensils	Light, malleable, good conductor of heat

Topic 4 Worksheet: Project

Using locally available materials (e.g. plasticine, matchsticks, bottle tops, beads), construct models to illustrate the bonding in: (a) sodium chloride (NaCl) (b) silicon(IV) oxide (SiO_2) (c) graphite (d) diamond. Label the atoms/ions and the type of bond in each model.

Topic 4 Exercise

1. Distinguish between ionic bonding and covalent bonding.
2. Draw a dot-and-cross diagram to show the bonding in a molecule of ammonia (NH_3) and in the ammonium ion (NH_4^+), clearly showing the dative bond.
3. Explain why graphite conducts electricity while diamond does not.
4. Explain why ionic compounds have high melting points.
5. State the type of intermolecular force present between water molecules and explain how it affects the boiling point of water.

Topic 4: End of Topic Questions

1. Explain, using electron arrangements, how an ionic bond is formed between calcium and chlorine to form calcium chloride, $CaCl_2$.
2. Distinguish between a single, double and triple covalent bond, giving an example of each.
3. Explain why metals are good conductors of electricity and heat, and why they are malleable and ductile.
4. Compare and contrast the properties of giant ionic and giant metallic structures.
5. Using bond type and structure, explain why aluminium is used to make aircraft bodies while silicon(IV) oxide is used to make abrasives.

TOPIC 5: PERIODICITY

By the end of this topic, you should be able to: describe the trends in physical properties of elements of the periodic table; investigate the chemical properties of elements in groups of the periodic table; describe the trends in properties across a period; and outline applications of elements of the periodic table.

5.1 What is Periodicity?

Periodicity refers to the repeating (periodic) trend in the physical and chemical properties of elements when they are arranged in order of increasing atomic number in the periodic table. These trends occur because of the regular, repeating pattern in electron arrangement as atomic number increases.

🔑 **KEY TERMS**

Atomic radius: a measure of the size of an atom, usually the distance from the nucleus to the outermost electron shell.

Ionisation energy: the minimum energy required to remove the most loosely held electron from an atom in its gaseous state.

Electron affinity: the energy change when an atom in its gaseous state gains an electron to form a negative ion.

Electronegativity: a measure of the ability of an atom to attract a shared pair of electrons in a covalent bond.

5.2 General Trends Down a Group

As you move down any group in the periodic table:

- Atomic radius increases – because more energy levels (shells) are added, moving the outer electrons further from the nucleus.
- Ionisation energy decreases – the outer electron is further from the nucleus and is shielded by more inner shells, so it is easier to remove.
- Metallic character increases – atoms lose electrons more easily.
- Melting and boiling points: generally decrease down Group I and II (metals become softer with weaker metallic bonding); but increase down Group VII (halogens, due to stronger Van der Waals forces in larger molecules).

5.3 General Trends Across a Period

As you move from left to right across a period (e.g. Period 3, from sodium to argon):

- Atomic radius decreases – the increasing number of protons pulls the electrons in the same shell more strongly towards the nucleus.
- Ionisation energy generally increases – it becomes harder to remove an electron as the nuclear attraction increases.
- Electronegativity increases – atoms attract shared electrons more strongly.
- Metallic character decreases while non-metallic character increases from left to right.

5.4 Group I: The Alkali Metals (Li, Na, K)

Physical properties: soft (can be cut with a knife), shiny when freshly cut (tarnish quickly in air), low density (Li, Na, K float on water), good conductors of heat and electricity, low melting and boiling points that decrease down the group.

Chemical properties: very reactive metals that become more reactive down the group, since it becomes progressively easier to lose the single outer electron.

Reaction	Observation / Product
With oxygen	Burn to form white metal oxides, e.g. $4Na + O_2 \rightarrow 2Na_2O$
With cold water	React vigorously, floating and fizzing, producing hydrogen gas and a metal hydroxide, e.g. $2Na + 2H_2O \rightarrow 2NaOH + H_2$
With chlorine gas	Burn to form white metal chlorides, e.g. $2Na + Cl_2 \rightarrow 2NaCl$
With dilute acids	React very vigorously (even explosively), producing hydrogen gas and a salt

⚠️ SAFETY FIRST

Alkali metals react violently with water and must only be handled by the teacher, using tweezers, in very small amounts, behind a safety screen.

Never touch alkali metals with bare hands.

5.5 Group II: The Alkaline Earth Metals (Mg, Ca)

Physical properties: harder and denser than Group I metals, higher melting points, still good conductors of heat and electricity.

Chemical properties: reactive, but less reactive than the corresponding Group I metal in the same period, since two electrons must be removed to form a stable ion. Reactivity increases down the group.

- Magnesium burns in oxygen with a brilliant white flame to form magnesium oxide (MgO).
- Calcium reacts steadily with cold water, while magnesium reacts only very slowly with cold water but readily with steam, producing hydrogen gas.
- Both react with dilute acids to produce hydrogen gas and a salt.

5.6 Group VII: The Halogens (F, Cl, Br, I)

Physical properties change gradually down the group:

Halogen	Colour	Physical State (room temp.)
Fluorine	Pale yellow	Gas
Chlorine	Greenish-yellow	Gas
Bromine	Reddish-brown	Liquid
Iodine	Grey-black (violet vapour)	Solid

Chemical properties: halogens are reactive non-metals, and reactivity decreases down the group (it becomes harder to gain an extra electron as atomic size increases).

- Chlorine dissolves in water to form chlorine water, which has bleaching properties due to the formation of chloric(I) acid (HOCl).
- Chlorine reacts with metals to form metal chlorides, e.g. $2Na + Cl_2 \rightarrow 2NaCl$.

- A more reactive halogen displaces a less reactive one from a solution of its salt, e.g. chlorine displaces bromine from potassium bromide solution: $Cl_2 + 2KBr \rightarrow 2KCl + Br_2$. This is called a displacement reaction and is used to compare the reactivity of halogens.

5.7 Group VIII (0): The Noble Gases (He, Ne, Ar)

Noble gases have a full outer electron shell, making them extremely stable and largely unreactive (inert). They exist as single atoms (monatomic) and have very low boiling points.

Uses: Helium is used to fill balloons and airships because it is lighter than air and non-flammable, unlike hydrogen. Neon is used in advertising signs because it glows when electricity passes through it. Argon is used to provide an inert atmosphere in light bulbs and during welding, to prevent the hot metal filament or weld from reacting with oxygen in the air.

🤔 REFLECT AND DISCUSS

1. Why do you think noble gases are used to preserve documents and artefacts in some museums?
2. In groups, discuss why alkali metals are stored under oil or kerosene rather than in the open air.

5.8 Trends Across Period 3 (Na to Ar)

Period 3 elements show a clear trend from metallic to non-metallic character:

Element	Na	Mg	Al	Si	P	S	Cl	Ar
Character	Metal	Metal	Metal	Metalloid	Non-metal	Non-metal	Non-metal	Noble gas

- Reaction with oxygen: Na, Mg and Al form basic or amphoteric oxides; Si, P, S form acidic oxides.
- Reaction with water: Na reacts vigorously with cold water; Mg reacts slowly with cold water but readily with steam; Al, Si, P, S, Cl do not react readily with cold water.
- Reaction with dilute acids: Na, Mg and Al react to liberate hydrogen gas; the non-metals generally do not react with dilute acids in this way.
- Melting points generally increase from Na to Si (giant structures) then drop sharply from P onwards (simple molecular structures).

📖 CASE STUDY: Chlorine: A Life-Saving Halogen

Chlorine gas is widely used to disinfect drinking water and swimming pools in Kenya and around the world. When dissolved in water, chlorine forms chloric(I) acid, which kills disease-causing microorganisms such as bacteria that cause cholera and typhoid.

However, chlorine gas is also highly toxic and must be handled with great care, using the correct safety equipment and concentrations, to protect both workers and the environment.

5.9 Applications of Elements Studied in Periodicity

Element	Application
Sodium	Sodium vapour street lamps; making soap (as sodium

Element	Application
	hydroxide)
Magnesium	Making lightweight alloys for aircraft; fireworks and flares (bright white flame)
Calcium	Making cement and lime (as calcium oxide/calcium carbonate); dietary supplements
Chlorine	Water treatment; making PVC plastics; bleaching agents
Argon	Inert atmosphere in light bulbs and welding

Topic 5 Worksheet

Design a practical investigation (with your teacher's guidance) to compare the reaction of magnesium ribbon and calcium metal with cold water. Record your observations in a table showing: (a) rate of bubbling (b) any heat produced (c) the gas produced and how you would test for it (d) your conclusion on the reactivity trend down Group II.

Topic 5 Exercise

1. State three physical properties of alkali metals.
2. Write a balanced equation for the reaction between potassium and cold water.
3. Explain why reactivity of Group I metals increases down the group.
4. Explain why reactivity of halogens decreases down the group.
5. State two uses each of chlorine and argon.

Topic 5: End of Topic Questions

1. Describe the trend in atomic radius (a) down a group (b) across a period, giving reasons for each trend.
2. Chlorine gas was bubbled through a solution of potassium iodide. State and explain the observation you would expect, and write a balanced equation for the reaction that occurs.
3. Compare the reactivity of sodium and magnesium with cold water, explaining your answer in terms of electron arrangement.
4. Explain why noble gases are generally unreactive.
5. Describe the trend in physical state and colour of the halogens from fluorine to iodine.

TOPIC 6: ACIDS AND BASES

By the end of this topic, you should be able to: explain the characteristics of acids and bases in aqueous solutions; describe the chemical properties of acids and bases; classify acids and bases into strong and weak; and outline the uses of acids and bases in day to day life.

6.1 What Are Acids and Bases?

An acid is a substance that dissolves in water to release hydrogen ions, H^+(aq) (more accurately, hydroxonium ions H_3O^+). A base is a substance that reacts with an acid to form a salt and water only. A base that is soluble in water is called an alkali, and it releases hydroxide ions, OH^-(aq), in solution.

Common laboratory acids include hydrochloric acid (HCl), sulphuric(VI) acid (H_2SO_4) and ethanoic acid (CH_3COOH). Common laboratory bases/alkalis include sodium hydroxide (NaOH), ammonia solution (NH_3) and sodium carbonate solution.

🔑 **KEY TERMS**

Acid: a substance that releases H^+ ions when dissolved in water.

Base: a substance that neutralises an acid to form a salt and water only.

Alkali: a soluble base that releases OH^- ions in aqueous solution.

Neutralisation: the reaction between an acid and a base to form a salt and water.

6.2 Chemical Properties of Acids

Acids react in characteristic ways with several classes of substances:

Reaction with	General Word Equation	Observation
Reactive metals	Acid + Metal → Salt + Hydrogen gas	Effervescence (fizzing); gas 'pops' with a lit splint
Metal carbonates	Acid + Carbonate → Salt + Water + Carbon dioxide gas	Effervescence; gas turns limewater milky/cloudy
Metal hydrogen carbonates	Acid + Hydrogen carbonate → Salt + Water + Carbon dioxide gas	Effervescence; gas turns limewater milky/cloudy
Metal oxides / hydroxides	Acid + Base → Salt + Water	Metal oxide/hydroxide dissolves/neutralised; no gas produced

Example equations:

- $Mg + 2HCl \rightarrow MgCl_2 + H_2$
- $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$
- $NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$
- $CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$
- $NaOH + HCl \rightarrow NaCl + H_2O$

⚠️ **SAFETY FIRST**

Always wear safety goggles and a lab coat when handling acids and bases.

Never taste any chemical in the laboratory.

In case of a spill on skin, wash immediately with plenty of clean water and inform your teacher.

Add acid to water, not water to acid, when diluting concentrated acids, to prevent dangerous spattering.

6.3 Indicators and the pH Scale

An indicator is a substance that changes colour depending on whether it is in an acidic or basic (alkaline) solution.

Indicator	Colour in Acid	Colour in Neutral	Colour in Base
Litmus	Red	Purple	Blue
Phenolphthalein	Colourless	Colourless	Pink
Methyl orange	Red	Orange	Yellow

The pH scale is a numerical scale, ranging from 0 to 14, that measures how acidic or alkaline a solution is. A universal indicator gives a range of colours corresponding to different pH values, allowing us to estimate the strength of an acid or base, not just whether it is acidic or basic.

pH Range	Nature of Solution	Universal Indicator Colour
0 – 3	Strongly acidic	Red
4 – 6	Weakly acidic	Orange/Yellow
7	Neutral	Green
8 – 10	Weakly alkaline	Blue
11 – 14	Strongly alkaline	Purple/Violet

💡 IDEA BOX: pH in Everyday Life

Human blood has a pH of about 7.35–7.45 (slightly alkaline), while gastric (stomach) juice has a pH of about 1.5–3.5 (strongly acidic) to help digest food and kill harmful microorganisms.

6.4 Strong and Weak Acids and Bases

Strength refers to the extent to which an acid or base ionises (dissociates) in water — this is different from concentration, which refers to the amount of acid or base dissolved in a given volume of water.

- A strong acid/base ionises almost completely in water, releasing a high concentration of H^+ or OH^- ions. Examples: hydrochloric acid, sulphuric(VI) acid, sodium hydroxide.
- A weak acid/base ionises only partially in water, releasing a lower concentration of H^+ or OH^- ions. Examples: ethanoic acid, ammonia solution, sodium carbonate solution.

Strong acids and bases have a lower/higher (further from 7) pH respectively than weak ones of the same concentration, and conduct electricity better than weak acids/bases of the same concentration, since they have more free ions in solution.

Solution	Type	Approx. pH (same concentration)	Electrical Conductivity
Hydrochloric acid	Strong acid	1	High
Ethanoic acid	Weak acid	3	Low
Sodium hydroxide	Strong base	13	High
Ammonia solution	Weak base	11	Low

🤔 REFLECT AND DISCUSS

1. Why do you think a dilute strong acid can sometimes have a similar pH to a concentrated weak acid?
2. Investigate: bring samples of common household substances (e.g. lemon juice, soap solution, vinegar) and predict whether each is acidic or basic before testing with litmus paper under your teacher's guidance.

6.5 Applications of Acids and Bases

Substance	Application
Sulphuric(VI) acid	Manufacture of fertilisers; car batteries; industrial cleaning
Hydrochloric acid	Present in gastric juice (digestion); cleaning metal surfaces (pickling)
Ethanoic acid	Main component of vinegar; food preservation
Sodium hydroxide	Making soap and detergents; paper manufacturing
Ammonia solution	Manufacture of fertilisers; household cleaning agents
Calcium hydroxide (lime)	Neutralising acidic soils in agriculture; water treatment

📖 CASE STUDY: Neutralising Acidic Soils

Many farms in parts of Kenya have naturally acidic soils, which limits the growth and yield of certain crops. Agricultural extension officers advise farmers to add agricultural lime (calcium hydroxide or calcium carbonate) to acidic soils. The lime neutralises the excess acidity, raising the soil pH to a level more suitable for crop growth, and also supplies calcium as a plant nutrient.

Topic 6 Worksheet

In the laboratory, under your teacher's supervision, carry out the following investigation. Record all observations in a table.

1. Add a few drops of universal indicator to separate samples of dilute hydrochloric acid, dilute ethanoic acid, sodium hydroxide solution and ammonia solution.
2. Record the colour observed and estimate the pH of each solution using the universal indicator colour chart.
3. Classify each solution as strong or weak, and acidic or basic.
4. Explain your classification.

Topic 6 Exercise

1. Define the terms acid, base and alkali.
2. Write balanced equations for the reaction between: (a) zinc and dilute sulphuric acid (b) sodium carbonate and dilute hydrochloric acid.
3. Distinguish between a strong acid and a concentrated acid.
4. State the colour change of phenolphthalein indicator when added to sodium hydroxide solution.
5. State two uses of sulphuric(VI) acid.

Topic 6: End of Topic Questions

1. Explain why a solution of hydrochloric acid conducts electricity better than a solution of ethanoic acid of the same concentration.
2. Describe an experiment you would carry out to distinguish between a strong base and a weak base of the same concentration.
3. Write a balanced equation for the reaction between copper(II) oxide and dilute sulphuric(VI) acid, and state the colour of the resulting solution.
4. A farmer notices that maize does not grow well on his farm. Soil tests show the soil is highly acidic. Explain how the farmer could improve the soil, and write an equation to support your answer.
5. Using the pH scale, explain how you would classify a solution with pH 5 and one with pH 9.

TOPIC 7: INTRODUCTION TO SALTS

By the end of this topic, you should be able to: classify different salts based on their properties; prepare salts using appropriate methods in the laboratory; describe the behaviour of salts when exposed to air; and outline applications of salts in day to day life.

7.1 What is a Salt?

A salt is a compound formed when the hydrogen ion(s) of an acid are wholly or partially replaced by a metal ion or an ammonium ion. Salts are usually formed during a neutralisation reaction between an acid and a base.

Common examples of salts include sodium chloride (table salt), calcium carbonate (found in limestone), and ammonium sulphate (a common fertiliser).

7.2 Types of Salts

Type of Salt	Description	Example
Normal salt	Formed when all the replaceable hydrogen ions of an acid are replaced by a metal or ammonium ion.	$NaCl$, $CuSO_4$, $(NH_4)_2SO_4$
Acidic salt	Formed when only some of the replaceable hydrogen ions of a polybasic acid are replaced.	$NaHSO_4$, $NaHCO_3$
Basic salt	Contains a hydroxide ion in addition to another anion.	$Pb(OH)Cl$, basic lead carbonate
Double salt	Formed when two different salts crystallise together in a fixed ratio from a mixed solution.	Potassium aluminium sulphate (potash alum), ammonium iron(II) sulphate

7.3 Solubility of Salts

Salts differ in their solubility in water. A general solubility guide for common anions is:

Anion	General Solubility Rule	Notable Exceptions (insoluble)
Chlorides	Mostly soluble	Silver chloride, lead(II) chloride
Sulphates	Mostly soluble	Barium sulphate, lead(II) sulphate, calcium sulphate (slightly soluble)
Nitrates	All soluble	None
Carbonates	Mostly insoluble	Sodium, potassium and ammonium carbonates are soluble

💡 IDEA BOX: A Handy Rule

All sodium, potassium and ammonium salts are soluble in water — this is a very useful rule to remember, since it applies regardless of which anion is attached!

7.4 Methods of Preparing Salts

The method used to prepare a salt in the laboratory depends mainly on its solubility.

(a) Preparing a Soluble Salt: Reaction of Acid with an Excess Metal, Insoluble Base or Carbonate

This method is used when the metal, base or carbonate is insoluble but the salt formed is soluble.

1. Add excess metal/insoluble base/carbonate to a warm, measured volume of dilute acid until no more reacts (effervescence stops/excess solid remains).
2. Filter off the excess unreacted solid to obtain a clear salt solution.
3. Evaporate the filtrate gently to concentrate it (evaporate to the point of crystallisation, indicated by a saturated solution).
4. Leave the concentrated solution to cool and crystallise.
5. Filter off the crystals and dry them between sheets of filter paper.

Example: $CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$

(b) Preparing a Soluble Salt: Titration (Acid + Soluble Base/Alkali)

This method is used when both the acid and the base are soluble (e.g. preparing sodium chloride from sodium hydroxide and hydrochloric acid), since there is no excess solid to filter off.

1. Titrate a known volume of the alkali against the acid using a suitable indicator to determine the exact volume of acid needed for complete neutralisation.
2. Repeat the reaction using the same volumes of acid and alkali but without the indicator, to obtain a pure salt solution.
3. Evaporate and crystallise the salt solution as before.

(c) Preparing an Insoluble Salt: Precipitation

This method is used to prepare an insoluble salt by mixing two suitable soluble salt solutions, so that the desired insoluble salt precipitates (settles) out of solution.

1. Mix appropriate volumes of two soluble salt solutions that will react to form the desired insoluble salt.
2. Filter the mixture to collect the precipitate (insoluble salt).
3. Wash the residue with distilled water to remove soluble impurities.
4. Dry the residue between sheets of filter paper or in an oven at low temperature.

Example (ionic equation): $Pb^{2+}(aq) + 2I^-(aq) \rightarrow PbI_2(s)$ — lead(II) iodide, a bright yellow precipitate, is formed when lead(II) nitrate solution is mixed with potassium iodide solution.

(d) Direct Synthesis (Combination)

Some salts can be prepared by directly combining a metal with a non-metal. Example: iron and sulphur are heated together to directly form iron(II) sulphide: $Fe(s) + S(s) \rightarrow FeS(s)$.

⚠️ SAFETY FIRST

Always heat solutions gently and evaporate salt solutions slowly to obtain good, well-formed crystals and to avoid spattering.

Wear safety goggles throughout the preparation and filtration process.

Dispose of excess chemicals and filtrates appropriately as directed by your teacher.

7.5 Behaviour of Salts When Exposed to Air

Term	Meaning	Example
Hygroscopic	A substance that absorbs moisture from the air but does not dissolve to form a solution.	Calcium chloride (anhydrous), silica gel
Deliquescent	A substance that absorbs so much moisture from the air that it dissolves completely, forming a solution.	Sodium hydroxide, calcium chloride (in humid conditions)
Efflorescent	A substance that loses its water of crystallisation to the air, forming a powdery surface.	Hydrated sodium carbonate (washing soda), $Na_2CO_3 \cdot 10H_2O$

🤔 REFLECT AND DISCUSS

1. Why do you think hygroscopic substances such as silica gel are packed together with electronics, shoes or medicines during transportation and storage?
2. In groups, design a simple experiment to demonstrate that a sample of washing soda crystals is efflorescent.

7.6 Applications of Salts

Salt	Application
Sodium chloride	Food seasoning and preservation; raw material for chlorine and sodium hydroxide manufacture
Ammonium sulphate / Ammonium nitrate	Nitrogenous fertilisers in agriculture
Calcium carbonate	Manufacture of cement, glass and lime; antacid tablets
Sodium hydrogen carbonate	Baking powder; antacid; fire extinguishers
Silver salts	Photographic film (light-sensitive silver halides)
Copper(II) sulphate	Fungicide in agriculture; electroplating

7.7 Environmental Impact of Salts: The Case of Fertilisers

While fertilisers such as ammonium nitrate and ammonium sulphate greatly improve crop yields, their overuse or misuse can cause serious environmental problems.

📖 CASE STUDY: Eutrophication: When Too Much of a Good Thing Becomes a Problem

When excess fertiliser is applied to farms near rivers and lakes, rainwater can wash the unused nutrients (nitrates and phosphates) into these water bodies. This process, called eutrophication, causes excessive growth of algae and water plants on the surface of the water.

As the algae eventually die and decompose, bacteria use up large amounts of dissolved oxygen in the water, leading to the death of fish and other aquatic organisms due to lack of oxygen. This has been observed in several water bodies in Kenya, including sections of Lake Victoria, where agricultural run-off has contributed to excessive water hyacinth growth.

Mitigation measures include: applying fertilisers in the correct recommended amounts, creating buffer zones of vegetation along rivers and lake shores to filter run-off, and practising soil testing before fertiliser application to avoid overuse.

🤔 REFLECT AND DISCUSS

1. As a young scientist, what would you advise farmers in your community to do to minimise the environmental effects of fertiliser use while still maintaining good crop yields?

Topic 7 Worksheet

Under your teacher's supervision, carry out a laboratory activity to prepare a sample of copper(II) sulphate crystals using the excess base/insoluble reactant method. Record: (a) the apparatus and reagents used (b) a step-by-step procedure (c) your observations (d) a labelled diagram of your set-up (e) the balanced equation for the reaction.

Topic 7 Exercise

1. Define the term 'salt' and give one example each of a normal salt, an acidic salt and a double salt.
2. State the general method used to prepare: (a) a soluble salt from an insoluble base (b) an insoluble salt.
3. Distinguish between hygroscopic, deliquescent and efflorescent substances, giving one example of each.
4. Write an ionic equation for the reaction between silver nitrate solution and sodium chloride solution to form insoluble silver chloride.
5. State three uses of sodium chloride.

Topic 7: End of Topic Questions

1. Describe, step by step, how you would prepare a pure, dry sample of zinc sulphate crystals in the laboratory, starting from zinc oxide and dilute sulphuric acid.
2. Explain why the titration method, rather than the excess solid method, is used to prepare sodium chloride from sodium hydroxide and hydrochloric acid.
3. A white solid was found to have turned into a puddle of liquid after being left in the open laboratory overnight. Name and explain this property, and suggest the identity of the class of substance the solid belongs to.
4. Explain the process of eutrophication and suggest two ways farmers can reduce its occurrence.
5. Classify the following as normal, acidic, basic or double salts: (a) Na_2SO_4 (b) $NaHSO_4$ (c) potassium aluminium sulphate (d) $CuCO_3$.

CHAPTER 8: ANSWER KEY

Use this chapter to check your work after attempting the Exercises and End of Topic Questions in each topic. Reflect and Discuss questions are open-ended and are not answered here — discuss these with your teacher and classmates.

Topic 1: Introduction to Chemistry — Exercise Answers

1. Chemistry is the branch of science that studies the composition, structure, properties and changes of matter.
2. Any four: Organic Chemistry (carbon compounds); Inorganic Chemistry (non-carbon substances, elements and minerals); Physical Chemistry (physical properties and energy changes); Analytical Chemistry (identifying/measuring composition); Biochemistry (chemistry of living things); Nuclear Chemistry (reactions in the nucleus).
3. Any three: Medicine/Pharmacy, Agriculture, Forensic science, Environmental science, Manufacturing/quality control, Teaching/research.
4. Drug abuse is the use of a drug without prescription, or in a way/dosage other than that prescribed by a qualified health worker, leading to harmful effects on the body and mind.
5. Any three: expiry/manufacture date, quality/certification mark (e.g. KEBS mark), nutritional information, preservatives used, cautionary/warning messages.

Topic 1: End of Topic Question Answers

1. Chemistry is called the 'central science' because it links and underpins many other sciences such as Biology, Physics, Geology and Agriculture, providing the basic understanding of matter needed in each of these fields.
2. Chemistry is applied in agriculture through the manufacture of fertilisers (to supply nutrients to crops) and pesticides/herbicides (to control pests and weeds), improving crop yields.
3. Any four: poor academic performance, damage to body organs (liver, kidney, brain), addiction, risky behaviour leading to accidents, breakdown of family/social relationships.
4. A prescription is an instruction from a qualified health worker on the correct drug and dosage to take. It should be followed strictly because taking the wrong drug, dosage or frequency can cause harm, ineffective treatment, or dangerous side effects.
5. Rights: right to a safe laboratory free of hazards; right to protective equipment. Responsibilities: following safety rules and instructions; reporting hazards or accidents immediately to the teacher.

Topic 2: The Atom — Exercise Answers

1. (a) Atomic number: the number of protons in the nucleus of an atom. (b) Mass number: the total number of protons and neutrons in the nucleus. (c) Isotope: atoms of the same element with the same atomic number but different mass numbers (different number of neutrons).
2. Protons = 12, Electrons = 12, Neutrons = 24 − 12 = 12.
3. R.A.M. = $[(79 \times 50.7) + (81 \times 49.3)] \div 100 = [4005.3 + 3993.3] \div 100 = 7998.6 \div 100 = 80.0$ (to 1 d.p.).
4. (a) Nitrogen: $1s^2\ 2s^2\ 2p^3$ (b) Silicon: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$ (c) Potassium: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$.
5. (a) An s orbital holds a maximum of 2 electrons. (b) A p sub-level holds a maximum of 6 electrons.

Topic 2: End of Topic Question Answers

1. Dalton proposed atoms as tiny, indivisible solid spheres. Thomson discovered the electron and proposed the 'plum pudding' model (positive sphere with embedded electrons). Rutherford's gold foil experiment showed atoms have

a small, dense, positively charged nucleus with electrons around it and mostly empty space. Bohr proposed that electrons occupy fixed circular energy levels around the nucleus.

2. Isotopes have the same number of protons and electrons, so they have the same electron arrangement, which determines chemical properties (how an atom reacts). Only the number of neutrons differs, which affects mass but not chemical behaviour.
3. R.A.M. = [(20 × 90.5) + (21 × 0.3) + (22 × 9.2)] ÷ 100 = [1810 + 6.3 + 202.4] ÷ 100 = 2018.7 ÷ 100 = 20.2 (to 1 d.p.).
4. A diagram showing a central nucleus (3 protons, 4 neutrons) with electrons arranged in two energy levels: 2 electrons in the first shell and 1 electron in the second shell (electron arrangement 2,1).
5. Element with atomic number 16 is Sulphur (S). Electron arrangement: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$.

Topic 3: The Periodic Table — Exercise Answers

1. A group is a vertical column of elements with the same number of valence electrons and similar chemical properties. A period is a horizontal row of elements with the same number of occupied energy levels (shells).
2. (a) Calcium chloride: $CaCl_2$ (b) Sodium sulphate: Na_2SO_4 (c) Ammonium nitrate: NH_4NO_3.
3. A diagram showing magnesium (2,8,2) losing 2 electrons (shown as dots) to two oxygen atoms (2,6), forming Mg^{2+} and two O^{2-} ions, each with a full outer shell (2,8 for Mg^{2+}; 2,8 for each O^{2-}).
4. (a) $4Na + O_2 \rightarrow 2Na_2O$ (b) $CaCO_3 \rightarrow CaO + CO_2$ (c) $2Fe + 3Cl_2 \rightarrow 2FeCl_3$.
5. In $FeCl_2$, iron has an oxidation number of +2. In $FeCl_3$, iron has an oxidation number of +3.

Topic 3: End of Topic Question Answers

1. Mendeleev arranged elements in order of increasing atomic mass and grouped them by similar properties, leaving gaps for undiscovered elements. Moseley later showed elements should be arranged by increasing atomic number instead, which corrected a few anomalies in Mendeleev's ordering (e.g. tellurium and iodine).
2. Electron arrangement 2,8,2 means: (a) Group II (2 valence electrons) (b) Period 3 (3 occupied shells) (c) It loses 2 electrons to form a 2+ ion (X^{2+}).
3. (a) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$ (b) $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$.
4. Elements in the same group have the same number of valence (outer shell) electrons, and it is these valence electrons that determine how an atom reacts, hence similar chemical properties.
5. Iron(III) ion is Fe^{3+}, sulphate ion is SO_4^{2-}. Crossing over valencies gives $Fe_2(SO_4)_3$.

Topic 4: Chemical Bonding — Exercise Answers

1. Ionic bonding involves the complete transfer of electrons from a metal to a non-metal atom, forming oppositely charged ions held by electrostatic attraction. Covalent bonding involves the sharing of electron pairs between two non-metal atoms.
2. In NH_3, nitrogen shares one electron pair with each of the three hydrogen atoms and has one lone pair. In NH_4^+, the lone pair on nitrogen is donated to an H^+ ion, forming a dative bond (shown with an arrow from N to the new H), giving nitrogen four bonds in total, three normal covalent and one dative.
3. Graphite conducts electricity because each carbon atom forms only 3 covalent bonds, leaving one delocalised electron per atom free to move and carry charge. Diamond does not conduct because each carbon atom forms 4 covalent bonds, leaving no free/delocalised electrons.
4. Ionic compounds have high melting points because of the strong electrostatic forces of attraction between oppositely charged ions throughout the giant ionic lattice, which require a large amount of energy to overcome.

5. Hydrogen bonding exists between water molecules (as H is bonded directly to the highly electronegative O). This significantly increases the boiling point of water compared to what would be expected for a molecule of its size, since extra energy is needed to break the hydrogen bonds between molecules.

Topic 4: End of Topic Question Answers

1. Calcium (2,8,8,2) loses 2 electrons to form Ca^{2+} (2,8,8). Each of two chlorine atoms (2,8,7) gains 1 electron to form Cl^- (2,8,8). The oppositely charged Ca^{2+} and $2Cl^-$ ions attract each other electrostatically to form $CaCl_2$.
2. Single bond: one shared electron pair, e.g. H–H in H_2. Double bond: two shared electron pairs, e.g. O=O in O_2. Triple bond: three shared electron pairs, e.g. N≡N in N_2.
3. Metals are good conductors because of the sea of delocalised electrons that can move freely and carry charge/heat energy. They are malleable and ductile because the layers of positive metal ions can slide over each other when force is applied, while still being held together by the surrounding sea of electrons, without breaking the structure.
4. Both giant ionic and giant metallic structures have high melting/boiling points due to strong forces holding the particles together. However, ionic solids do not conduct electricity (ions are fixed in place) but conduct when molten/dissolved, while metals conduct electricity in both solid and molten states due to free delocalised electrons. Ionic solids are typically brittle, while metals are malleable and ductile.
5. Aluminium is light, malleable and a good conductor of heat due to its giant metallic structure, making it ideal for aircraft bodies. Silicon(IV) oxide has a giant covalent (atomic) structure with strong covalent bonds throughout, making it extremely hard, hence its use in abrasives.

Topic 5: Periodicity — Exercise Answers

1. Any three: soft (can be cut with a knife), shiny when freshly cut, low density, good conductors of heat/electricity, low melting/boiling points.
2. $2K + 2H_2O \rightarrow 2KOH + H_2$
3. Reactivity increases down Group I because the outer electron is further from the nucleus and more shielded by inner shells in larger atoms, so it is more easily lost.
4. Reactivity decreases down Group VII because the atoms get larger, so it becomes harder for the nucleus to attract and gain an additional electron.
5. Chlorine: water treatment/disinfection; manufacture of PVC. Argon: inert atmosphere in light bulbs; inert atmosphere for welding.

Topic 5: End of Topic Question Answers

1. (a) Down a group, atomic radius increases because more energy levels (shells) are added. (b) Across a period, atomic radius decreases because the increasing nuclear charge pulls the outer electrons (in the same shell) closer to the nucleus.
2. The solution turns from colourless to brown/orange as iodine is displaced (chlorine is more reactive than iodine and displaces it from solution). Equation: $Cl_2 + 2KI \rightarrow 2KCl + I_2$.
3. Sodium is more reactive than magnesium with cold water because sodium loses its single outer electron more easily than magnesium, which must lose two electrons. Sodium reacts vigorously with cold water while magnesium reacts only very slowly with cold water.
4. Noble gases have a full outer electron shell, giving them a stable electron arrangement, so they have no tendency to lose, gain or share electrons, making them largely unreactive.
5. Fluorine and chlorine are pale yellow and greenish-yellow gases respectively; bromine is a reddish-brown liquid; iodine is a grey-black solid (that gives off a violet vapour when heated). Colour deepens and physical state changes from gas to liquid to solid down the group.

Topic 6: Acids and Bases — Exercise Answers

1. Acid: a substance that releases H^+ ions in aqueous solution. Base: a substance that neutralises an acid to form a salt and water only. Alkali: a soluble base that releases OH^- ions in aqueous solution.
2. (a) $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$ (b) $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$.
3. A strong acid is one that ionises almost completely in water, regardless of its concentration. A concentrated acid is one that has a large amount of acid dissolved in a small volume of water (little water present) — concentration and strength are independent properties.
4. Phenolphthalein turns from colourless to pink when added to sodium hydroxide solution (a base).
5. Any two: manufacture of fertilisers, car batteries, industrial cleaning/pickling of metals.

Topic 6: End of Topic Question Answers

1. Hydrochloric acid, being a strong acid, ionises almost completely in water, producing a high concentration of free H^+ and Cl^- ions to carry charge. Ethanoic acid, a weak acid, ionises only partially, producing fewer free ions, hence lower conductivity.
2. Measure equal volumes and concentrations of the strong base and weak base. Test the electrical conductivity of each using a conductivity apparatus (bulb and electrodes) — the strong base will conduct more strongly (brighter bulb) than the weak base, since it produces more free ions in solution.
3. $CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$. The resulting solution is blue, due to the formation of copper(II) sulphate.
4. The farmer should add agricultural lime (calcium hydroxide/calcium carbonate) to the soil to neutralise the excess acidity. Equation: $Ca(OH)_2 + 2HX \rightarrow CaX_2 + 2H_2O$ (where HX represents the acidic substance in the soil), or $CaCO_3$ neutralising the acid similarly.
5. A solution with pH 5 is weakly acidic (pH below 7). A solution with pH 9 is weakly alkaline/basic (pH above 7).

Topic 7: Introduction to Salts — Exercise Answers

1. A salt is a compound formed when the hydrogen ion(s) of an acid are wholly or partially replaced by a metal or ammonium ion. Normal salt: NaCl. Acidic salt: $NaHCO_3$. Double salt: potassium aluminium sulphate.
2. (a) Reaction of the acid with excess insoluble base/metal/carbonate, followed by filtration, evaporation and crystallisation. (b) Precipitation — mixing two suitable soluble salt solutions, then filtering, washing and drying the precipitate.
3. Hygroscopic: absorbs moisture from the air without dissolving (e.g. anhydrous calcium chloride). Deliquescent: absorbs so much moisture that it dissolves completely (e.g. sodium hydroxide). Efflorescent: loses its own water of crystallisation to the air (e.g. hydrated sodium carbonate).
4. $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$; ionic equation: $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$.
5. Any three: food seasoning, food preservation, raw material for manufacturing chlorine/sodium hydroxide.

Topic 7: End of Topic Question Answers

1. Add zinc oxide in small portions to warm, measured dilute sulphuric acid until it is in excess (no more dissolves). Filter to remove excess zinc oxide. Gently evaporate the filtrate to the point of crystallisation. Leave to cool and crystallise. Filter off the zinc sulphate crystals and dry between filter paper. Equation: $ZnO + H_2SO_4 \rightarrow ZnSO_4 + H_2O$.
2. Since both sodium hydroxide and hydrochloric acid are soluble, there is no excess solid to filter off at the end of the reaction, so titration is used to measure the exact volumes needed for complete neutralisation, ensuring a pure salt solution without excess acid or alkali.

3. This is deliquescence — the solid absorbed moisture from the air until it dissolved completely to form a solution. The solid likely belongs to a class of hygroscopic/deliquescent salts, such as sodium hydroxide or certain metal chlorides.
4. Eutrophication occurs when excess nutrients (nitrates/phosphates from fertiliser run-off) enter water bodies, causing excessive algae growth. When the algae die and decompose, bacteria consume large amounts of dissolved oxygen, killing fish and other aquatic life. Farmers can reduce this by applying fertiliser in recommended amounts only, and by maintaining vegetation buffer zones along rivers and lakes to filter run-off.
5. (a) Na_2SO_4 — normal salt (b) $NaHSO_4$ — acidic salt (c) potassium aluminium sulphate — double salt (d) $CuCO_3$ — normal salt.